Phase 2 Model Performance Report

AlphaEarth to SAR Cross-Modal Prediction

Full dataset size: 2880 rows

Balanced modeling subset: 720 rows

Regions: amazon_forest, california_coast, iowa_ag, sf_bay_urban

Dynamic World classes: 9

Held-out test split: 70/30 stratified by region x dw_label

CV tuning: StratifiedGroupKFold grouped by spatial blocks within region

Models compared: ridge regression vs tuned LightGBM

Best held-out model by target:

- S1_VH: Ridge regression with R2=0.940, RMSE=1.163, MAE=0.902, r=0.970
- S1_VV: Ridge regression with R2=0.893, RMSE=1.253, MAE=0.921, r=0.945
- S1_VV_div_VH: Tuned LightGBM with R2=0.719, RMSE=1.038, MAE=0.791, r=0.850

Key interpretation:

- Tuned LightGBM improved over the old fixed-tree baseline and clearly beats ridge on VV/VH ratio (0.
- Ridge still remains best on S1_VV (0.893 vs 0.867) and S1_VH (0.940 vs 0.919).
- S1_VH remains the easiest target overall; the VV/VH ratio remains the hardest.

Overall Held-out Performance

target	model	cv_r2_mean	cv_rmse_mean	r2	rmse	mae	pearson_r	best_params
S1_VH	Ridge regression	0.921	1.297	0.940	1.163	0.902	0.970	{"alpha": 10.0}
S1_VH	Tuned LightGBM	0.894	1.498	0.919	1.352	1.013	0.959	{"colsample_bytree": 0.5038788416660758, "
S1_VV	Ridge regression	0.862	1.361	0.893	1.253	0.921	0.945	{"alpha": 10.0}
S1_VV	Tuned LightGBM	0.843	1.471	0.867	1.395	0.998	0.933	{"colsample_bytree": 0.5779972601681014, "
S1_VV_div_VH	Tuned LightGBM	0.598	1.126	0.719	1.038	0.791	0.850	{"colsample_bytree": 0.5644075875290285, "
S1_VV_div_VH	Ridge regression	0.595	1.097	0.698	1.076	0.798	0.837	{"alpha": 10.0}

Metrics are computed on the 30 percent held-out test set. cv_r2_mean and cv_rmse_mean come from grouped CV on the training split.

Regional S1_VV Performance

region	model	r2	rmse	mae	pearson_r	n_test
amazon_forest	Tuned LightGBM	0.936	1.100	0.839	0.969	54
amazon_forest	Ridge regression	0.945	1.018	0.859	0.974	54
california_coast	Tuned LightGBM	0.890	1.206	0.889	0.946	54
california_coast	Ridge regression	0.883	1.246	0.901	0.943	54
iowa_ag	Tuned LightGBM	0.768	1.219	0.957	0.883	54
iowa_ag	Ridge regression	0.865	0.930	0.732	0.936	54
sf_bay_urban	Tuned LightGBM	0.792	1.908	1.305	0.892	54
sf_bay_urban	Ridge regression	0.839	1.680	1.191	0.916	54

Each region contributes 54 held-out samples in the balanced test split.

Land-Use S1_VV Performance

dw_label_name	model	r2	rmse	mae	pearson_r	n_test
bare	Tuned LightGBM	0.586	2.063	1.466	0.773	24
bare	Ridge regression	0.746	1.617	1.093	0.871	24
built	Tuned LightGBM	0.715	1.944	1.467	0.902	24
built	Ridge regression	0.724	1.910	1.328	0.879	24
crops	Tuned LightGBM	0.832	0.911	0.760	0.914	24
crops	Ridge regression	0.887	0.747	0.616	0.946	24
flooded_vegetation	Tuned LightGBM	0.957	0.764	0.579	0.984	24
flooded_vegetation	Ridge regression	0.939	0.909	0.761	0.969	24
grass	Tuned LightGBM	0.746	1.151	0.869	0.868	24
grass	Ridge regression	0.733	1.179	0.913	0.880	24
shrub_and_scrub	Tuned LightGBM	0.600	1.527	1.031	0.802	24
shrub_and_scrub	Ridge regression	0.857	0.912	0.805	0.929	24
snow_and_ice	Tuned LightGBM	0.894	1.130	0.904	0.954	24
snow_and_ice	Ridge regression	0.955	0.736	0.663	0.981	24
trees	Tuned LightGBM	0.581	1.409	1.097	0.763	24
trees	Ridge regression	0.476	1.575	1.235	0.714	24
water	Tuned LightGBM	0.921	1.074	0.806	0.964	24
water	Ridge regression	0.916	1.110	0.874	0.965	24

Subgroup diagnostics for the primary S1_VV target on the held-out test set.

Feature Importance Summary

Top 5 embedding dimensions by target and model

S1_VV

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  Ridge: A27 (0.950), A34 (0.865), A17 (0.735), A31 (0.707), A44 (0.696)
  LightGBM: A27 (0.149), A25 (0.121), A31 (0.057), A19 (0.050), A34 (0.033)
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S1_VH

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  Ridge: A27 (1.362), A34 (0.863), A53 (0.793), A44 (0.766), A63 (0.751)
  LightGBM: A25 (0.262), A27 (0.095), A63 (0.050), A49 (0.028), A34 (0.025)
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S1_VV_div_VH

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  Ridge: A12 (0.614), A25 (0.491), A62 (0.429), A27 (0.412), A03 (0.406)
  LightGBM: A25 (0.166), A63 (0.102), A24 (0.050), A07 (0.035), A30 (0.031)
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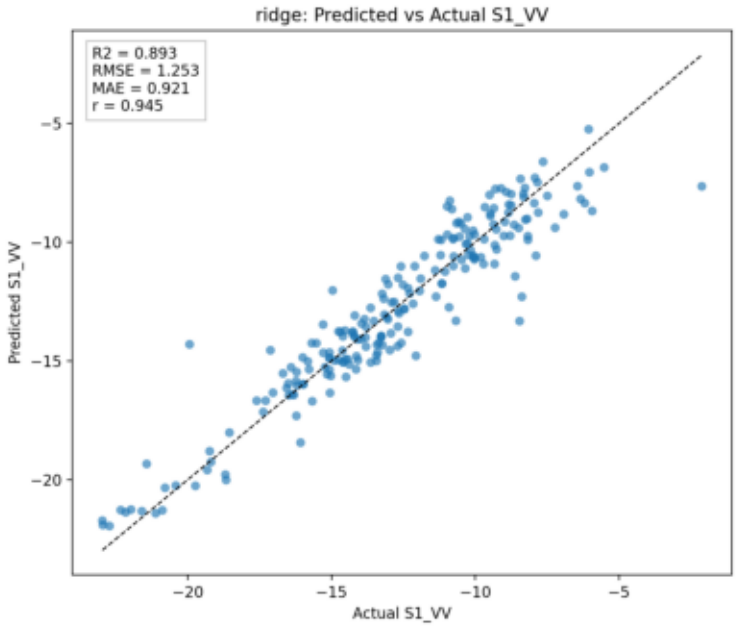
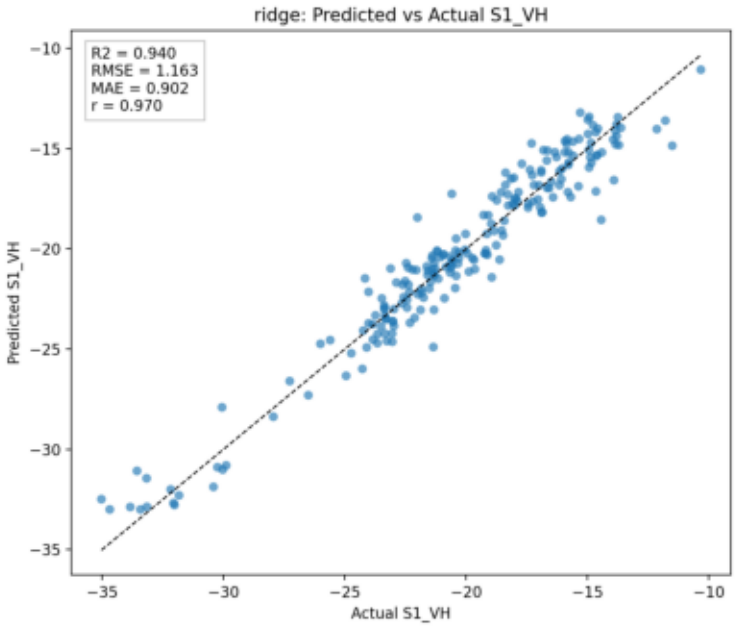
Predicted vs Actual: Ridge Regression

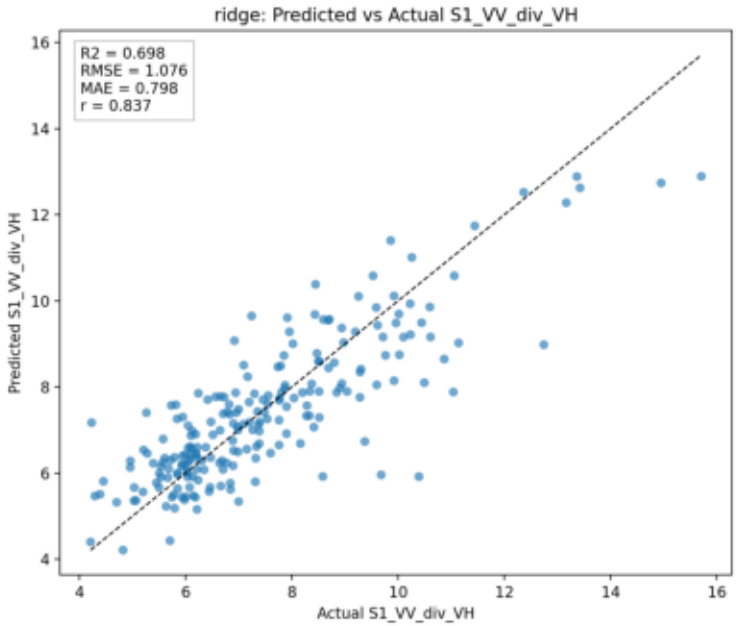
predicted vs actual S1 VV ridge



predicted vs actual S1 VH ridge

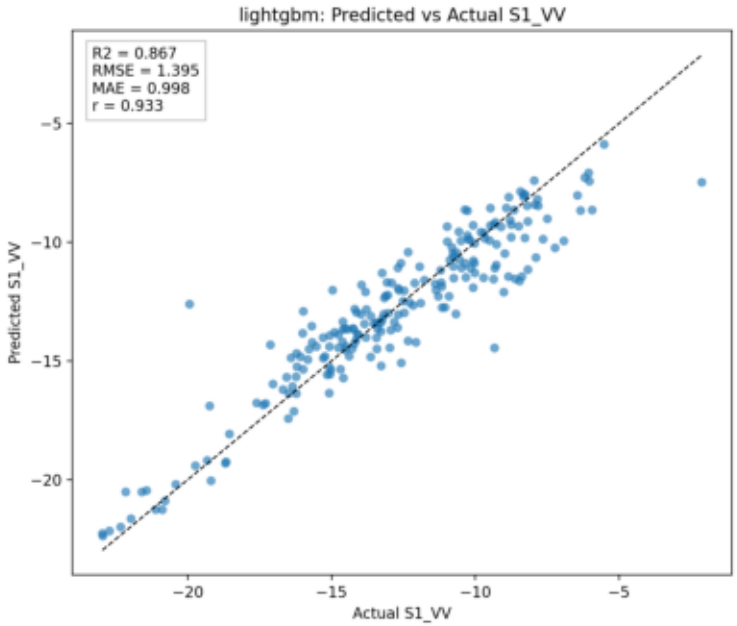


predicted vs actual S1 VV div VH ridge

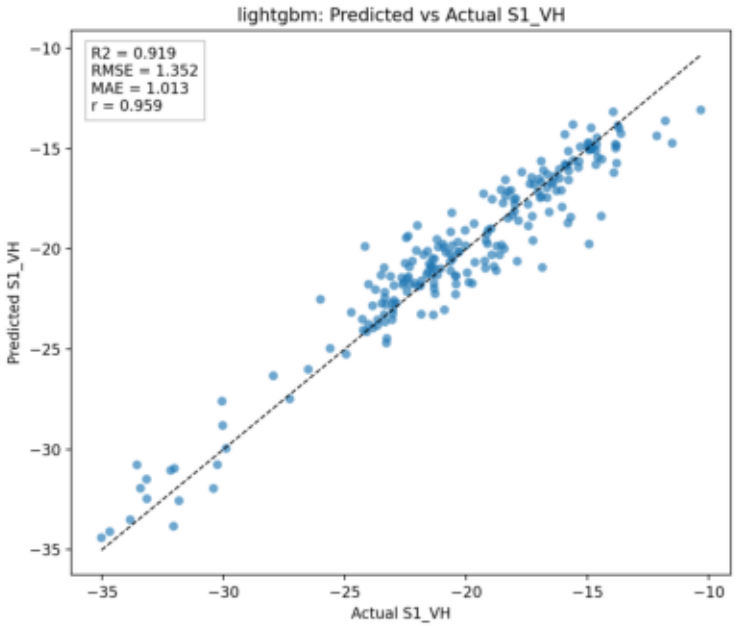


Predicted vs Actual: Tuned LightGBM

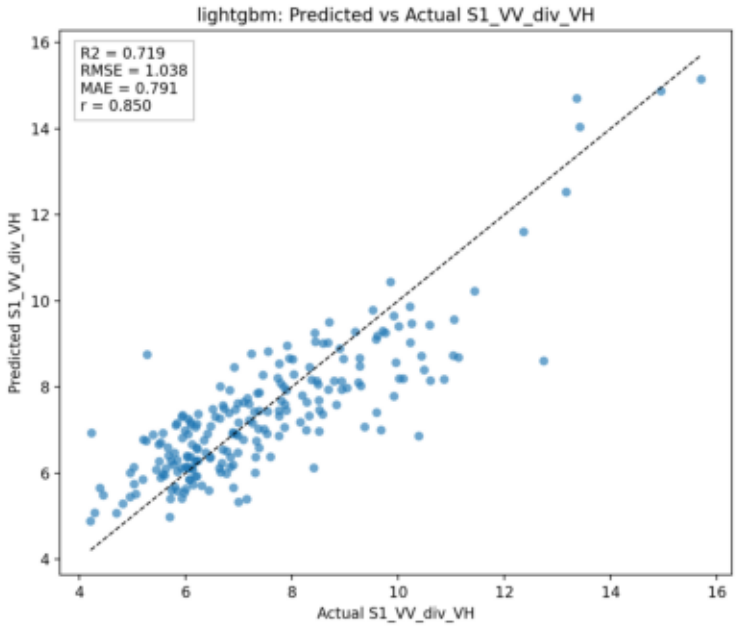
predicted vs actual S1 VV lightgbm



predicted vs actual S1 VH lightgbm

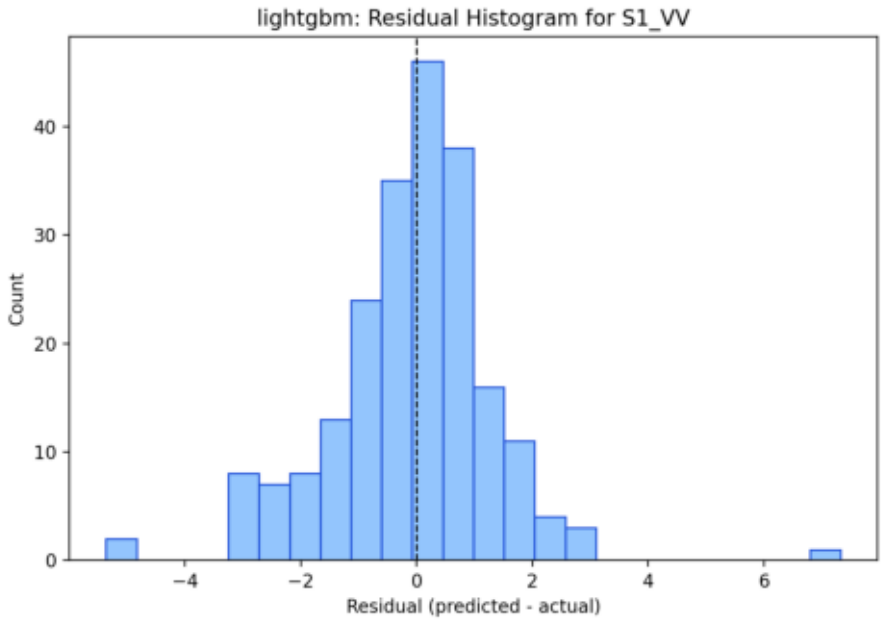


predicted vs actual S1 VV div VH lightgbm

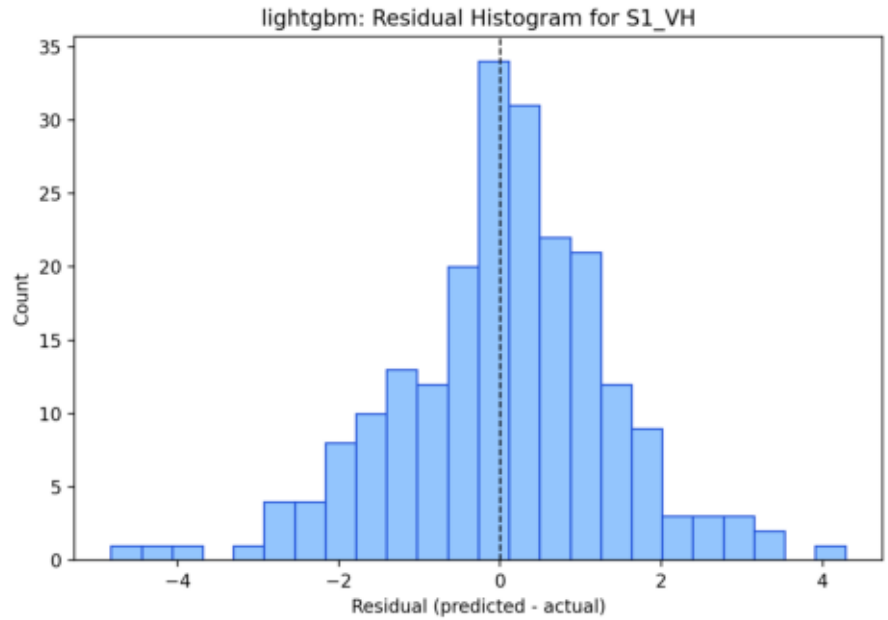
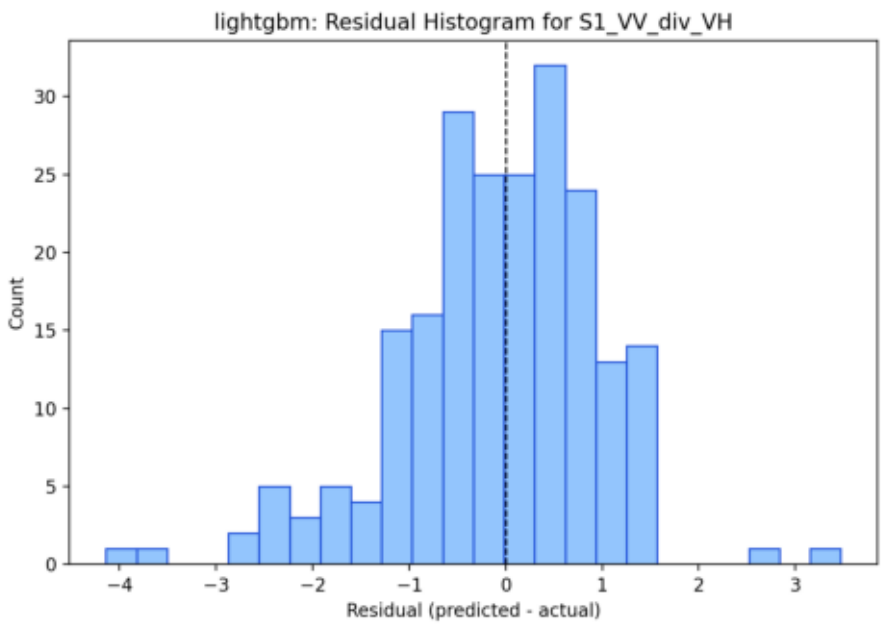


Residual Histograms: Tuned LightGBM

residual histogram S1 VV lightgbm



residual histogram S1 VH lightgbm



residual histogram S1 VV div VH lightgbm